



CSE 308 Operating System Sessional Project Ideas :

Course Code: CSE 308 | Course Title: Operating System Sessional

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Common Extension Hub



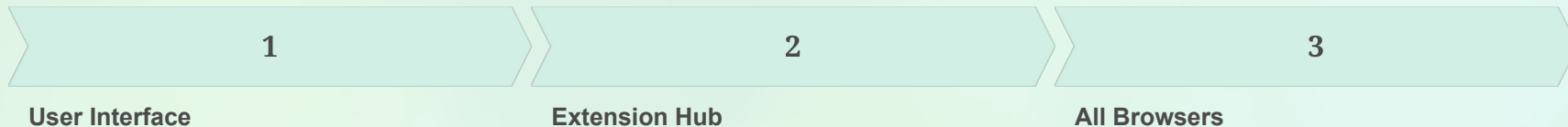
Problem

Installing extensions separately in Chrome, Edge, Firefox, and Brave is **repetitive** and difficult to manage.

Goal

A single application to **search, install, update, enable/disable, and remove** extensions across all browsers.

Architecture Flow



Common Extension Hub

OS Concepts

**Processes &
Process
Management**

**Inter-Process
Communication
(IPC)**

**File System &
Browser Profiles**

**User Permissions
& Access Control**

**Threading &
Concurrency**

Implementation

- Desktop application + browser profile detection + extension management module
- **Expected:** Single-click extension management across all browsers

Evaluation Metrics

- Installation time & user steps
- Resource usage & usability feedback

Future Scope

- Cloud sync, enterprise deployment
- Extension recommendations, Linux package integration

CompanionCat — Intelligent Linux Desktop Companion

Problem

Users notice high CPU, RAM, temperature, or battery issues **too late**.

Goal

Real-time Linux monitoring through an **animated desktop cat companion**.

Monitored Metrics

- CPU, RAM, Disk, Network, Battery, Temperature, Process Count
- Taskbar icon & smart alerts

Cat Expressions



Happy

System healthy



Sleeping

Low activity



Working

Moderate load



Angry

High CPU/RAM



Scared

Critical alert



Hungry

Low battery

CompanionCat

OS Concepts



Process Management & Scheduling



Threads & Synchronization



System Monitoring
(/proc, /sys)



Signal Handling & IPC



Resource Management

Implementation

- Monitoring daemon + UI widget + tray icon + alert engine
- Expected: Lightweight always-visible system companion

Evaluation Metrics

- CPU overhead, memory usage, detection latency
- UI responsiveness & user study

Future Scope

- Historical graphs, remote monitoring
- Anomaly detection, mobile notifications

CommandAgent — Human Language Linux Command Assistant



Problem

New Linux users **struggle to remember** terminal commands.

Goal

Execute Linux tasks from **simple human-language instructions**.

Example Commands

- “Show large files”
- “Create a folder named Project”
- “Check memory usage”
- “Kill Chrome”

Execution Flow



CommandAgent

OS Concepts

Process Creation & Execution

Shell & Command Interpretation

Inter-Process Communication (IPC)

File System & Permissions

Threading & Concurrency

Implementation

- Rule-based command parser + safe execution engine + command whitelist + logging
- **Expected:** Beginner-friendly Linux command assistant

Evaluation Metrics

- Command accuracy & execution success rate
- Response time & usability testing

Future Scope

- Voice input, multilingual commands
- Scripting mode, learning-based suggestions

Thank You

Questions & Discussion

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